Denosumab

Denosumab is used to strengthen and protect the bones from the effect of cancer or cancer treatment.

On this page

- What is denosumab?
- How denosumab is given
- About side effects
- Side effects while treatment is being given
- Very common side effects
- Other side effects
- Other important information
- How we can help

What is denosumab?

Denosumab is used to strengthen and protect the bones from the effect of cancer or cancer treatment. It is a type of targeted therapy drug called a monoclonal antibody.

Denosumab can be given in slightly different ways. The way you have it will depend on your situation.

It may be given to prevent bone problems, such as fractures, if you have cancer that has spread to the bones (secondary bone cancer).

It may be given if you have prostate cancer and have weakened bones due to hormonal therapy. You might also have it if you have breast cancer and have weakened bones after going through the menopause.

It is best to read this information with our general information about targeted therapy drugs and the type of cancer you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

How denosumab is given

You usually have denosumab as an outpatient. Sometimes, you, or someone who looks after you, may be shown how to give the injections so you can have them at home.

You have denosumab as an injection under the skin (subcutaneously). You have the injection in the thigh, stomach or upper arm. You may have this:

- every 4 weeks
- every 3 months
- every 6 months.

You will have regular blood tests to check the amount of calcium in your blood. This is to check that it is safe for you to have treatment.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Your course of treatment

Your cancer team will discuss your treatment plan with you. They will explain how often you will get the injections.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having this treatment or shortly after they have it.

Allergic reaction

Rarely, some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

Low calcium in the blood

You will have regular blood tests to check your calcium levels. If they are low, your doctor, nurse or pharmacist will prescribe calcium and vitamin D supplements for you.

Contact your cancer team or hospital straight away if you get any of these signs:

- muscle spasms, twitching or cramps
- tingling around your mouth or in your fingers and toes
- drowsiness or confusion.

Muscle, bone or joint pain

This treatment can cause pain in your muscles, bones or joints. This may be more common in your hands and feet. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better.

A rare side effect of this treatment is a fracture of the thigh bone (femur). Tell your doctor, nurse or pharmacist if you have any new or unusual pain in your hip, groin or thigh during or after treatment.

Breathlessness

This treatment may make you feel breathless. Always tell your doctor, nurse or pharmacist if this happens or if you have any trouble with your breathing.

Diarrhoea

This treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

Jaw problems (osteonecrosis)

Osteonecrosis of the jaw is when healthy bone tissue becomes damaged and dies.

The risk of getting osteonecrosis can increase due to:

- gum disease
- problems with dentures
- some dental treatments.

Your doctor, nurse or pharmacist will advise you to have a full dental check-up before starting denosumab. It is important to them if you are having or plan to have dental treatment. If you have any mouth ulcers or sores, you should not start treatment until they have healed.

It is very important to look after your teeth during and after treatment with denosumab. Brush them regularly and have regular dental check-ups. Let your dentist know you are having denosumab.

Some of the symptoms of osteonecrosis can include:

- pain and swelling in your jaw
- redness of the gums or gum infections
- loosening of teeth.

Tell your cancer team and dentist straight away if you have any of these symptoms. Rarely, osteonecrosis can also affect the bones around the ears. If you have any ear pain, infection, or fluid leaking (discharge) from your ear, tell your doctor, nurse or pharmacist straight away.

Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice or prescribe creams or medicines to help.

Occasionally this treatment can cause a skin infection. If you notice an area of skin becomes red, swollen and warm to touch, tell your doctor, nurse or pharmacist straight away.

Hair loss

This treatment can cause your hair to get thinner. If you notice this, tell your nurses or pharmacist. Your nurse can give you advice and support on hair care.

Constipation

This treatment can cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- Eat high-fibre foods, such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Tummy pain

This treatment can cause pain in your tummy (abdomen). Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Infections

You may have more urinary tract infections or chest infections. Tell your doctor, nurse or pharmacist if you have any symptoms linked with infection.

Back pain, tingling or numbness down your leg (sciatica)

This treatment can cause back pain and sciatica. Tell your doctor, nurse or pharmacist if this happens. They can give you painkillers to help.

Contact your cancer team or the hospital straight away if you have:

- numbness or tingling in your lower back or legs - this may be caused by pressure on nerves in or around the spine
- weakness or loss of sensation in your legs
- loss of bladder or bowel control.

Low levels of phosphate in the blood

This treatment can cause low levels of phosphate in the blood. Phosphate is a mineral. Phosphates have different functions in the body including keeping your bones and teeth healthy.

You will have regular blood tests to check phosphate levels. Your doctor may prescribe medicine to take if your phosphate levels are low.

Sweating more than usual

Some people may find that they sweat more than usual while having this treatment. Try to drink plenty of fluids if this happens. You can talk to your doctor, nurse or pharmacist if you have any concerns.

Second tumour

In people with advanced cancer this treatment is linked to a risk of developing another type of tumour. This could be non-cancerous or cancer. Your doctor, nurse or pharmacist can give you more information.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a blood clot. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Sorbitol (a type of sugar)

This treatment contains sorbitol. Tell your doctor, nurse or pharmacist if you think you have an intolerance to some sugars. They can advise you if denosumab is safe for you to take.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and coronavirus (covid). They may also recommend other vaccines, such as Shingrix® for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026

Date reviewed

Reviewed: 01 August 2025 | Next review: 01 August 2027

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/denosumab · retrieved 19 October 2026